

Streaming Temporal Kernels for Dynamic Regret with Logarithmic State Complexity: Proofs

Supplementary proof document

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A Supplementary definitions

The prescribed density and its integer-sampled kernel are

$$Q(\omega) = \frac{c_Q}{|\omega| \left(1 + \frac{|\omega|^2}{4\pi^2}\right)^{1/4} \nu(\omega) [\ln \nu(\omega)]^2}, \quad \nu(\omega) = \ln(\pi + |\omega|^{-1/2}), \quad c_Q = \frac{1}{4} \ln \ln \pi. \quad (1)$$

$$k^Q(s, t) = \int_{\mathbb{R}} Q(\omega) e^{-2\pi i(s-t)\omega} d\omega. \quad (2)$$

For $J(\theta) = \lfloor \log_2 |\theta|^{-1} \rfloor$, define

$$H(\theta) = \begin{cases} 1 + 2^{J(\theta)} a_{J(\theta)}, & 0 < |\theta| \leq 1, \\ 1, & \text{otherwise,} \end{cases} \quad q_Q(\theta) = \sum_{n \in \mathbb{Z}} Q(n + \frac{\theta}{2\pi}). \quad (3)$$

The finite magnitude update used in the regret proof is

$$x_t = \varepsilon_0 \sum_{i=0}^{I_t} \pi_i \eta_i e^{\eta_i s_t - \eta_i^2 q_t}, \quad \pi_i = \frac{1}{(i+1)(i+2)}, \quad \eta_i = \frac{2^{-i}}{2G}, \quad I_t = \lceil 2 \log_2(t+1) \rceil, \quad \varepsilon_0 = \frac{3}{4} \varepsilon. \quad (4)$$

Writing $I = \lceil \frac{1}{2} \log_2 T \rceil$, let

$$\mathcal{L}(r, T) = 1 + \ln \left(1 + \frac{8rG}{3\varepsilon} 2^I (I+1)(I+2) \right). \quad (5)$$

B Proofs

Throughout, $w(x) = 1/\lceil x(\ln x)^2 \rceil$, $\lambda_j = 2^{-j}$, $\rho_j = e^{-\lambda_j}$ and $R_{\rho, T} = (\rho^{|s-t|})_{s,t=1}^T$.

Lemma 1. *Let $K \in \mathbb{R}^{T \times T}$ be symmetric and strictly positive definite. The inner product $\langle f, h \rangle_{\mathcal{H}_K} = f^\top K^{-1} h$ makes $\mathbb{R}^T$ an RKHS with sections Ke_t and $\langle f, Ke_t \rangle_{\mathcal{H}_K} = f_t$. For every $u \in \mathbb{R}^T$,*

$$u^\top K^{-1} u = \sup_{x \in \mathbb{R}^T} \{2u^\top x - x^\top K x\}. \quad (6)$$

If $cA \preceq B \preceq CA$ with A, B strictly positive definite and $0 < c \leq C$, then $C^{-1} \mathcal{E}_A(U) \leq \mathcal{E}_B(U) \leq c^{-1} \mathcal{E}_A(U)$. If $\varphi_1, \dots, \varphi_T$ in a real Hilbert space satisfy $\langle \varphi_s, \varphi_t \rangle = K_{st}$, then $u^\top K^{-1} u$ is the least squared norm among all v with $\langle v, \varphi_t \rangle = u_t$ for every t .

Proof of Lemma 1. Since K^{-1} is symmetric and strictly positive definite, $\langle f, h \rangle_{\mathcal{H}_K} = f^\top K^{-1} h$ is an inner product, and the space has dimension T , hence is complete. For any f , $\langle f, K e_t \rangle_{\mathcal{H}_K} = f^\top K^{-1} K e_t = f^\top e_t = f_t$, which is the reproducing property. Evaluation is continuous because Cauchy Schwarz gives $|f_t| \leq \|f\|_{\mathcal{H}_K} \|K e_t\|_{\mathcal{H}_K} = \|f\|_{\mathcal{H}_K} \sqrt{K_{tt}}$. This proves the RKHS claim.

For the variational identity, expand the square

$$(x - K^{-1}u)^\top K (x - K^{-1}u) = x^\top K x - 2u^\top x + u^\top K^{-1}u.$$

The left side is nonnegative and vanishes at $x = K^{-1}u$, so $2u^\top x - x^\top K x \leq u^\top K^{-1}u$ with equality attained, which is (6).

For the energy comparison, suppose $B \succeq cA$. For every x , $2u^\top x - x^\top B x \leq 2u^\top x - cx^\top A x$, and taking the supremum over x on the right, using (6) for the matrix cA , yields $u^\top B^{-1}u \leq c^{-1}u^\top A^{-1}u$. From $B \preceq CA$ we get $A \succeq C^{-1}B$; applying the inequality just proved to this pair gives $u^\top A^{-1}u \leq C u^\top B^{-1}u$. Applying both bounds to each column of U and summing gives the energy comparison.

For the interpolation claim, set $\alpha = K^{-1}u$ and $v_\star = \sum_s \alpha_s \varphi_s$. Then $\langle v_\star, \varphi_t \rangle = \sum_s \alpha_s K_{st} = (K\alpha)_t = u_t$ and $\|v_\star\|^2 = \alpha^\top K \alpha = u^\top K^{-1}u$. If v also interpolates, put $r = v - v_\star$; then $\langle r, \varphi_t \rangle = 0$ for each t , hence $\langle r, v_\star \rangle = \sum_t \alpha_t \langle r, \varphi_t \rangle = 0$ and $\|v\|^2 = \|v_\star\|^2 + \|r\|^2 \geq \|v_\star\|^2$. $\square$

Proposition 1. *There are strictly positive definite matrices $K_\delta, \tilde{K}_\delta$ whose entrywise distance tends to 0 while the ratio of $u^\top \tilde{K}_\delta^{-1}u$ to $u^\top K_\delta^{-1}u$ diverges. Moreover, if $\tilde{K}$ has rank less than T , no $c > 0$ satisfies $cK \preceq \tilde{K}$.*

Proof of Proposition 1. For $0 < \delta < 1$ take $K_\delta = \text{diag}(1, \delta)$ and $\tilde{K}_\delta = \text{diag}(1, \delta^2)$. Both are strictly positive definite and the largest entrywise difference is $\delta - \delta^2 \leq \delta \rightarrow 0$. For $u = (0, 1)^\top$, $u^\top K_\delta^{-1}u = \delta^{-1}$ while $u^\top \tilde{K}_\delta^{-1}u = \delta^{-2}$, so the ratio is $1/\delta \rightarrow \infty$. For the rank claim, if $\text{rank } \tilde{K} < T$ there is $x \neq 0$ with $\tilde{K}x = 0$; then $cK \preceq \tilde{K}$ would give $0 < cx^\top K x \leq x^\top \tilde{K} x = 0$, a contradiction. $\square$

Lemma 2. *The weights (4) satisfy $a_j = \int_{j+e}^{j+e+1} \frac{dx}{x(\ln x)^2} > 0$, $\sum_{j \geq 0} a_j = 1$ and $\sum_{j > J} a_j = 1/\ln(J+e+1)$. With $w(x) = 1/[x(\ln x)^2]$ one has $w(j+e+1) \leq a_j \leq w(j+e)$, the sequence a_j is non-increasing, and for $0 \leq r \leq J$,*

$$\frac{a_{J-r}}{a_J} \leq (r+2)^3, \quad \sum_{j=0}^J 2^j a_j \leq 102 \cdot 2^J a_J. \quad (7)$$

Proof of Lemma 2. The derivative of $x \mapsto 1/\ln x$ is $-w(x)$, so the fundamental theorem of calculus gives $a_j = \int_{j+e}^{j+e+1} w$, and the integrand is strictly positive because the interval lies in $(1, \infty)$. Telescoping, $\sum_{j=0}^L a_j = 1 - 1/\ln(L+e+1)$, which tends to 1 and also yields the stated tail mass. Next, $w'(x) = -\frac{\ln x + 2}{x^2(\ln x)^3} < 0$ for $x \geq e$, so a unit-length integral lies between the values at the endpoints, giving $w(j+e+1) \leq a_j \leq w(j+e)$; shifting the interval to the right cannot increase the integral, so a_j is non-increasing.

For $b \geq a \geq e$ the derivative of $x \mapsto \ln x/x$ is $(1 - \ln x)/x^2 \leq 0$, whence $\ln b/\ln a \leq b/a$. Taking $a = J - r + e$ and $b = J + e + 1$,

$$\frac{a_{J-r}}{a_J} \leq \frac{w(J-r+e)}{w(J+e+1)} = \frac{b}{a} \left(\frac{\ln b}{\ln a} \right)^2 \leq \left(\frac{b}{a} \right)^3,$$

and $b/a = 1 + \frac{r+1}{J-r+e} \leq 1 + \frac{r+1}{e} \leq r+2$, which is the first part of (7).

For the geometric moment put $S_k = \sum_{r \geq 0} r^k 2^{-r}$, so $S_0 = 2$ and, for $k \geq 1$, $S_k = \frac{1}{2} \sum_{r \geq 0} (r+1)^k 2^{-r}$. Solving successively, $S_1 = 2$, $S_2 = 6$ and $S_3 = 26$. Hence $\sum_{r \geq 0} (r+2)^3 2^{-r} = S_3 + 6S_2 + 12S_1 + 8S_0 = 102$. Substituting $r = J - j$,

$$\sum_{j=0}^J 2^j a_j = 2^J a_J \sum_{r=0}^J 2^{-r} \frac{a_{J-r}}{a_J} \leq 102 \cdot 2^J a_J. \quad \square$$

Lemma 3. For $0 < \rho < 1$ the kernel $R_\rho(s, t) = \rho^{|s-t|}$ has discrete spectral density $p_\rho(\theta) = \frac{1-\rho^2}{1-2\rho \cos \theta + \rho^2}$ and $R_{\rho,T} \succeq \frac{1-\rho}{1+\rho} I_T$. Put $p_- = 2e^{-2}$, $d_0 = \min\{e^{-2}, 4e^{-1}/\pi^2\}$ and $p_+ = 2/d_0$. If $0 < \lambda \leq 1$ and $\rho = e^{-\lambda}$, then

$$p_- \frac{\lambda}{\lambda^2 + \theta^2} \leq p_\rho(\theta) \leq p_+ \frac{\lambda}{\lambda^2 + \theta^2}. \quad (8)$$

Consequently $K_T^\infty \succeq \beta I_T$ for every T .

Proof of Lemma 3. The absolutely convergent geometric series gives

$$1 + \sum_{h \geq 1} \rho^h (e^{ih\theta} + e^{-ih\theta}) = \frac{1 - \rho^2}{1 - 2\rho \cos \theta + \rho^2}.$$

Since $\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{in\theta} d\theta$ equals 1 for $n = 0$ and 0 otherwise, multiplying the series by $e^{i(s-t)\theta}$ and integrating termwise, which is legitimate by uniform convergence, returns $\rho^{|s-t|}$. The same computation gives, for $x \in \mathbb{R}^T$,

$$x^\top R_{\rho,T} x = \frac{1}{2\pi} \int_{-\pi}^{\pi} p_\rho(\theta) \left| \sum_t x_t e^{it\theta} \right|^2 d\theta.$$

The denominator of p_ρ is at most $(1 + \rho)^2$, so $p_\rho \geq \frac{1-\rho}{1+\rho}$, and expanding the squared modulus yields $R_{\rho,T} \succeq \frac{1-\rho}{1+\rho} I_T$.

For (8), the integral representation of the exponential gives $e^{-1}\lambda \leq 1 - e^{-\lambda} \leq \lambda$ and $2e^{-2}\lambda \leq 1 - e^{-2\lambda} \leq 2\lambda$ for $0 < \lambda \leq 1$. On $|\theta| \leq \pi$ we have $\frac{2\theta^2}{\pi^2} \leq 1 - \cos \theta \leq \frac{\theta^2}{2}$, the left inequality because $\sin$ is concave on $[0, \pi/2]$ and therefore lies above its chord. Writing the denominator as $(1 - e^{-\lambda})^2 + 2e^{-\lambda}(1 - \cos \theta)$ and using $e^{-\lambda} \geq e^{-1}$,

$$d_0(\lambda^2 + \theta^2) \leq (1 - e^{-\lambda})^2 + 2e^{-\lambda}(1 - \cos \theta) \leq \lambda^2 + \theta^2,$$

and combining with the bounds on the numerator gives (8).

Finally each mixture component is positive semidefinite, and the component $j = 0$ has weight a_0 and factor e^{-1} , so

$$x^\top K_T^\infty x \geq a_0 x^\top R_{e^{-1},T} x \geq \beta \|x\|^2.$$

The infinite sum is legitimate because every kernel entry is at most 1 in absolute value and the weights sum to 1. $\square$

Lemma 4. The kernel k^∞ has integrable density $f(\theta) = \sum_{j \geq 0} a_j p_{\rho_j}(\theta)$, normalized by $\frac{1}{2\pi} \int_{-\pi}^{\pi} f = 1$. With $c_f = \frac{1}{2} \min\{\beta, p_-/2\}$ and $C_f = 106 p_+$,

$$c_f H(\theta) \leq f(\theta) \leq C_f H(\theta) \quad a.e. \quad (9)$$

Proof of Lemma 4. Each p_{ρ_j} is nonnegative with normalized integral 1, and the integral of a series of nonnegative functions is the increasing limit of the partial-sum integrals, so $\frac{1}{2\pi} \int f = \sum_j a_j = 1$ and f is integrable; the kernel representation follows because $|e^{i\theta}| = 1$ and f dominates the partial sums.

Let $0 < |\theta| \leq 1$ and $J = J(\theta)$, so $2^{-(J+1)} < |\theta| \leq 2^{-J} = \lambda_J$. The J -th component satisfies $\frac{\lambda_J}{\lambda_J^2 + \theta^2} \geq \frac{1}{2\lambda_J} = 2^{J-1}$, so $f(\theta) \geq \frac{p_-}{2} 2^J a_J$; Lemma 3 also gives $f(\theta) \geq \beta$. Adding the two lower bounds and halving gives the lower bound in (9). For the upper bound, $j \leq J$ implies $\frac{\lambda_j}{\lambda_j^2 + \theta^2} \leq 2^j$, so Lemma 2 bounds the coarse scales by $102 \cdot 2^J a_J$; while for $j > J$, using $a_j \leq a_J$ and $\theta^{-2} < 2^{2J+2}$,

$$\sum_{j>J} a_j \frac{\lambda_j}{\lambda_j^2 + \theta^2} \leq 2^{2J+2} a_J \sum_{j>J} 2^{-j} = 4 \cdot 2^J a_J.$$

Adding and multiplying by p_+ gives $f \leq 106 p_+ 2^J a_J \leq C_f H$. If $1 < |\theta| \leq \pi$ then every $\frac{\lambda_j}{\lambda_j^2 + \theta^2} \leq 1$, so $f \leq p_+$, while $f \geq \beta$; the constants in the statement cover this band. $\square$

Lemma 5. *Q is integrable, the integer-sampled kernel has discrete density $q_Q(\theta) = \sum_{n \in \mathbb{Z}} Q(n + \frac{\theta}{2\pi})$, and there are explicit constants $c_Q, C_Q > 0$ with $c_Q H \leq q_Q \leq C_Q H$ almost everywhere.*

Proof of Lemma 5. Write $c_Q = \frac{1}{4} \ln \ln \pi$ and set

$$\begin{aligned} \alpha_0 &= \min\{\frac{1}{2}, \frac{\ln 2}{2}, \frac{\ln(2\pi)}{2e}\}, & \alpha_1 &= \max\{1, \frac{\ln 2}{2}, \frac{\ln(\pi + \sqrt{4\pi})}{e}\}, \\ y_0 &= \ln(\pi + \sqrt{2\pi}), & v_0 &= \max\{e, \alpha_0^{-2}\}, \\ \alpha_2 &= \min\{\frac{1}{2}, \frac{\ln y_0}{\ln v_0}\}, & \alpha_3 &= 1 + \ln \alpha_1, \end{aligned}$$

$A_{\text{sh}} = (1 + \frac{1}{16\pi^4})^{1/4}$, $C_w = (1 + e^{-1})^3$, $L_Q = \frac{2\pi c_Q}{A_{\text{sh}} \alpha_1 \alpha_3^2}$, $U_Q = \frac{4\pi c_Q C_w}{\alpha_0 \alpha_2^2}$, $H_Q = \frac{c_Q \sqrt{2\pi}}{\ln \pi (\ln \ln \pi)^2}$, $A_Q = 6 \cdot 2^{3/2} H_Q$, $B_Q = \frac{2\pi c_Q}{\ln \pi (\ln \ln \pi)^2}$ and $b_Q = c_Q [\frac{3}{2}(1 + \frac{9}{16\pi^2})^{1/4} \ln(\pi + \sqrt{2}) [\ln \ln(\pi + \sqrt{2})]^2]^{-1}$. We prove the claim with $c_q = \frac{1}{2} \min\{b_Q, L_Q\}$ and $C_q = \max\{U_Q, A_Q + B_Q\}$.

To control Q on dyadic bands, let $0 < \theta \leq 1$ and let $j \geq 0$ be the unique integer with $2^{-(j+1)} < \theta \leq 2^{-j}$. Put $x = \theta/(2\pi)$, $v = j + e$ and $y = \ln(\pi + x^{-1/2})$. From $x \leq 2^{-j}/(2\pi)$,

$$y \geq \ln(x^{-1/2}) \geq \frac{j}{2} \ln 2 + \frac{1}{2} \ln(2\pi) \geq \alpha_0 v,$$

the last inequality because $\alpha_0 \leq \frac{\ln 2}{2}$ and $e\alpha_0 \leq \frac{\ln(2\pi)}{2}$. From $x > 2^{-(j+1)}/(2\pi)$ we get $\pi + x^{-1/2} \leq (\pi + \sqrt{4\pi}) 2^{j/2}$, hence $y \leq \alpha_1 v$; and $x \leq 1/(2\pi)$ forces $y \geq y_0 > 1$. To control $\ln y$: if $v \geq v_0$ then $v \geq \alpha_0^{-2}$, so $\ln y \geq \ln v + \ln \alpha_0 \geq \frac{1}{2} \ln v$; if $e \leq v \leq v_0$ then $\ln y \geq \ln y_0 \geq \frac{\ln y_0}{\ln v_0} \ln v$. Both cases give $\ln y \geq \alpha_2 \ln v$, while $y \leq \alpha_1 v$ and $\ln v \geq 1$ give $\ln y \leq \alpha_3 \ln v$. Since also $2\pi 2^j \leq x^{-1} < 4\pi 2^j$ and $1 \leq (1 + \frac{x^2}{4\pi^2})^{1/4} \leq A_{\text{sh}}$, substituting into (1) yields

$$\frac{2\pi c_Q}{A_{\text{sh}} \alpha_1 \alpha_3^2} 2^j w(v) \leq Q(x) \leq \frac{4\pi c_Q}{\alpha_0 \alpha_2^2} 2^j w(v). \quad (10)$$

Lemma 2 gives $a_j \leq w(v)$ and $a_j \geq w(v+1)$, and

$$\frac{w(v)}{w(v+1)} = \frac{v+1}{v} \left(\frac{\ln(v+1)}{\ln v} \right)^2 \leq (1 + e^{-1})^3 = C_w,$$

so $a_j \leq w(v) \leq C_w a_j$ and (10) becomes $L_Q 2^j a_j \leq Q(\theta/2\pi) \leq U_Q 2^j a_j$. As Q is even, the same holds for negative θ .

The same estimate proves integrability. On the j -th positive band $(\frac{2^{-(j+1)}}{2\pi}, \frac{2^{-j}}{2\pi}]$ the upper bound is $U_Q 2^j a_j$ and the length is $2^{-(j+1)}/(2\pi)$, so the band integral is at most $U_Q a_j/(4\pi)$; summing over j and using $\sum_j a_j = 1$ shows Q is integrable on $(0, \frac{1}{2\pi}]$, and evenness handles the negative side. For any $x > 0$ we have $(1 + \frac{x^2}{4\pi^2})^{1/4} \geq \sqrt{\frac{x}{2\pi}}$ while the two logarithmic factors are at least $\ln \pi$ and $(\ln \ln \pi)^2$, so

$$Q(x) \leq H_Q x^{-3/2}, \quad (11)$$

which is integrable away from the origin.

For the aliased density, partition $\mathbb{R}$ into $n + [-\frac{1}{2}, \frac{1}{2})$ and substitute $\omega = n + \xi$. Because $s - t$ is an integer, $e^{-2\pi i(s-t)n} = 1$, and absolute integrability permits exchanging the sum with the integral:

$$\begin{aligned} k^Q(s, t) &= \int_{-1/2}^{1/2} \sum_{n \in \mathbb{Z}} Q(n + \xi) e^{-2\pi i(s-t)\xi} d\xi \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} q_Q(\theta) e^{-i(s-t)\theta} d\theta, \end{aligned}$$

which also shows q_Q is integrable.

To bound the remaining bands, let $|\theta| \leq \pi$ and $\xi = \theta/(2\pi)$, so $|\xi| \leq \frac{1}{2}$. For $n \neq 0$, $|n + \xi| \geq |n| - \frac{1}{2} \geq \frac{|n|}{2}$, so by (11), $\sum_{n \neq 0} Q(n + \xi) \leq 2H_Q 2^{3/2} \sum_{n \geq 1} n^{-3/2}$. Since $n^{-3/2} \leq \int_{n-1}^n x^{-3/2} dx$ for $n \geq 2$, the series is at most $1 + \int_1^\infty x^{-3/2} dx = 3$, so the non-zero aliases contribute at most A_Q . On the other hand $1 + \xi \in [\frac{1}{2}, \frac{3}{2}]$, and on that interval every positive factor in the denominator of (1) is at most the corresponding factor in the definition of b_Q , so

$$q_Q(\theta) \geq Q(1 + \xi) \geq b_Q. \quad (12)$$

If $1 < |\theta| \leq \pi$ then $|\xi| > \frac{1}{2\pi}$, so $|\xi|^{-1} < 2\pi$ and the global lower bounds on the logarithmic factors give $Q(\xi) \leq B_Q$; hence $b_Q \leq q_Q \leq A_Q + B_Q$ on that band. If $0 < |\theta| \leq 1$, the dyadic estimate and (12) give

$$q_Q(\theta) \geq \max\{b_Q, L_Q 2^J a_J\} \geq \frac{1}{2} \min\{b_Q, L_Q\} (1 + 2^J a_J),$$

and $q_Q(\theta) \leq A_Q + U_Q 2^J a_J \leq C_q (1 + 2^J a_J)$. All bands are covered. $\square$

Theorem 1. *With $c_\star = c_f/C_q$ and $C_\star = C_f/c_q$, the comparison (5) holds for every $T \geq 1$, and both matrices are strictly positive definite.*

Proof of Theorem 1. Lemmas 4 and 5 give, almost everywhere,

$$\begin{aligned} f &\geq c_f H \geq \frac{c_f}{C_q} q_Q = c_\star q_Q, \\ f &\leq C_f H \leq \frac{C_f}{c_q} q_Q = C_\star q_Q. \end{aligned}$$

For $x \in \mathbb{R}^T$ put $X(\theta) = \sum_{t=1}^T x_t e^{it\theta}$. The spectral representations give

$$x^\top K_T^\infty x = \frac{1}{2\pi} \int f |X|^2, \quad x^\top K_T^Q x = \frac{1}{2\pi} \int q_Q |X|^2.$$

Since $|X|^2 \geq 0$, multiplying the pointwise density inequalities by $|X|^2$ and integrating yields $c_\star x^\top K_T^Q x \leq x^\top K_T^\infty x \leq C_\star x^\top K_T^Q x$ for every x , which is (5). Finally (12) gives $K_T^Q \succeq b_Q I_T$, so both matrices are strictly positive definite. $\square$

Proof of Theorem 1. For positive definiteness and the unit diagonal, fix j and a finite T , let $e_1, \dots, e_T$ be orthonormal, set $\varphi_{j,1} = e_1$ and, for $t \geq 2$,

$$\begin{aligned} r_{j,t} &= e^{-\lambda_j(\sigma_j(t) - \sigma_j(t-1))} \in (0, 1], \\ \varphi_{j,t} &= r_{j,t} \varphi_{j,t-1} + \sqrt{1 - r_{j,t}^2} e_t. \end{aligned}$$

By induction $\varphi_{j,t-1}$ involves only the first $t-1$ basis vectors, so it is orthogonal to e_t and $\|\varphi_{j,t}\|^2 = 1$. For $s < t$, repeated substitution gives $\langle \varphi_{j,s}, \varphi_{j,t} \rangle = \prod_{r=s+1}^t r_{j,r} = e^{-\lambda_j(\sigma_j(t) - \sigma_j(s))}$, since the internal clock is non-decreasing. Thus each component is a Gram matrix, hence positive semidefinite; the weighted sum converges entrywise and is positive semidefinite, and the component $j = 0$ has $n_{0,\delta} = 1$, so its internal time difference equals the external one and it is the strictly positive definite $R_{e^{-1},T}$. Every component equals 1 on the diagonal and the weights sum to 1, so $\bar{k}^\delta(t, t) = 1$.

We next bound the error from delaying one component. Fix $\lambda \in (0, 1]$ and an integer $n \geq 1$, and compare $R(s, t) = e^{-\lambda|s-t|}$ with $R^{[n]}(s, t) = e^{-\lambda|(t-n)_+ - (s-n)_+|}$. Split $[T]$ into $A = \{t \leq n\}$ and $B = \{t > n\}$. If both indices lie in B the difference vanishes. If both lie in A the difference is $1 - e^{-\lambda|s-t|}$, of modulus at most λn , so the early block contributes at most

$$\lambda n \left(\sum_{s \in A} |x_s| \right)^2 \leq \lambda n^2 \|x_A\|^2. \quad (13)$$

If $s \in A$ and $t \in B$ the difference factors exactly as

$$e^{-\lambda(t-n)} - e^{-\lambda(t-s)} = (1 - e^{-\lambda(n-s)}) e^{-\lambda(t-n)},$$

a rank-one matrix $pq^\top$ with $p_s = 1 - e^{-\lambda(n-s)}$ and $q_t = e^{-\lambda(t-n)}$. Then $\|p\|^2 \leq n(\lambda n)^2 = \lambda^2 n^3$ and, using $e^{2\lambda} - 1 \geq 2\lambda$, $\|q\|^2 \leq \sum_{h \geq 1} e^{-2\lambda h} = \frac{1}{e^{2\lambda} - 1} \leq \frac{1}{2\lambda}$. Hence the cross block and its transpose contribute

$$\begin{aligned} |2(x_A^\top p)(q^\top x_B)| &\leq \|p\| \|q\| (\|x_A\|^2 + \|x_B\|^2) \\ &\leq \sqrt{\lambda/2} n^{3/2} \|x\|^2, \end{aligned} \quad (14)$$

where the first inequality uses $(\|x_A\| - \|x_B\|)^2 \geq 0$. Adding (13) and (14),

$$|x^\top (R_T^{[n]} - R_T)x| \leq (\lambda n^2 + \sqrt{\lambda/2} n^{3/2}) \|x\|^2, \quad (15)$$

a bound with no factor growing in T .

To sum over delayed components, write $J = J_{0,\delta}$. Components with $j \leq J$ are unchanged. For $j = J + r$ with $r \geq 1$ we have $n_{j,\delta} \leq 2 \cdot 2^{r/4}$ and $\lambda_j = 2^{-J-r}$, so

$$\lambda_j n_{j,\delta}^2 \leq 4 \cdot 2^{-J} 2^{-r/2}, \quad \sqrt{\lambda_j/2} n_{j,\delta}^{3/2} \leq 2 \cdot 2^{-J/2} 2^{-r/8}.$$

Multiplying (15) by $a_j \leq 1$, summing and evaluating the two geometric series gives $|x^\top (\bar{K}_T^\delta - K_T^\infty)x| \leq (D_1 2^{-J} + D_2 2^{-J/2}) \|x\|^2$. The definition of $J_{0,\delta}$ gives $2^{-J/2} \leq \frac{\delta\beta}{D_1 + D_2}$, and $J \geq 0$ gives $2^{-J} \leq 2^{-J/2}$, so the bracket is at most $\delta\beta$. Since $K_T^\infty \succeq \beta I_T$ by Lemma 3, the perturbation is at most $\delta x^\top K_T^\infty x$, and separating the two sides of the absolute value gives (8). The bound does not use T , so one kernel serves all prefixes.

For the active count, if $t \geq 1$ is an integer and $j > J$, then $\lceil 2^{(j-J)/4} \rceil \leq t$ iff $2^{(j-J)/4} \leq t$ iff $j \leq J + 4 \log_2 t$; adding the always-active indices $0, \dots, J$ gives the stated count. Finally (10) follows by applying Lemma 1 to (8) composed with (5), which together give $(1 - \delta)c_\star K_T^Q \preceq \bar{K}_T^\delta \preceq (1 + \delta)C_\star K_T^Q$. $\square$

Lemma 6. *For any gradient stream, the identities (9) hold, together with $N_{t+1} = N_t + 2\langle g_t, h_t \rangle + \|g_t\|^2$; a component activating at round $t + 1$ has exact state $z_{j,t+1} = p_{t+1}$; and at most four components activate per round. Hence the interface (3) is maintained exactly in time and space $O(d[1 + \log(t + 1) + \log(1/\delta)])$ without storing past gradients.*

Proof of Lemma 6. For an active index $j \leq J(t)$ define $z_{j,t} = \sum_{s < t} e^{-\lambda_j(\sigma_j(t) - \sigma_j(s))} g_s$. If $j > J(t)$ then $n_{j,\delta} > t$, so every $s \leq t$ has $\sigma_j(s) = 0$ and the entire inactive tail contributes exactly $A_{J(t)} p_t$ at round t ; this gives the expression for h_t in (9).

For an already active component, $t \geq n_{j,\delta}$ implies $\sigma_j(t + 1) - \sigma_j(t) = 1$, so expanding the finite sum at round $t + 1$ gives $z_{j,t+1} = \rho_j(z_{j,t} + g_t)$. For a component first activating at round $t + 1$ we have $n_{j,\delta} = t + 1$, hence $\sigma_j(s) = \sigma_j(t + 1) = 0$ for every $s \leq t$, and therefore

$$z_{j,t+1} = \sum_{s \leq t} g_s = p_{t+1}, \quad p_{t+1} = p_t + g_t.$$

This is the exact state of the new component, not a reconstruction of a forgotten exponentially weighted history, precisely because the component's clock had not been running.

For the norm statistic, split the double sum defining N_{t+1} according to whether both indices are smaller than t , exactly one equals t , or both equal t . The first group is N_t ; symmetry of the kernel makes the second $2 \sum_{s < t} k(t, s) \langle g_t, g_s \rangle = 2 \langle g_t, h_t \rangle$; and the third is $k(t, t) \|g_t\|^2 = \|g_t\|^2$.

Finally, the number of active states is $J(t) + 1$, and from round t to $t + 1$ at most four components activate, because $4 \log_2(t + 1) - 4 \log_2 t = 4 \log_2(1 + 1/t) \leq 4$ and the floors of two reals differing by at most 4 differ by at most 4. Each state occupies d reals and each update costs $O(d)$ arithmetic operations, giving the stated complexity; no past gradient is retained. $\square$

Lemma 7. *Extending $u_{1:T}$ by zero and setting $\widehat{U}(\theta) = \sum_{t=1}^T u_t e^{it\theta}$,*

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} |\theta| \|\widehat{U}(\theta)\|^2 d\theta \leq \frac{M(P_T + 2M)(1 + \ln T) + M^2}{c_b}. \quad (16)$$

Proof of Lemma 7. We use a nonnegative series that dominates $|\theta|$. Put $B(\theta) = \sum_{h \geq 1} \frac{1 - \cos(h\theta)}{h^2}$, which converges since each term is nonnegative and at most $2/h^2$. Let $x = |\theta|$. If $1 \leq x \leq \pi$, keeping the first term and using $1 - \cos x \geq \frac{2x^2}{\pi^2}$ gives $B(\theta) \geq \frac{2x^2}{\pi^2} \geq \frac{2x}{\pi^2}$. If $0 < x \leq 1$, let $n = \lceil 1/x \rceil$; for $h = n, \dots, 2n - 1$ we have $1 \leq hx \leq 2 + x \leq 3 < \pi$, and since $\cos$ is non-increasing on $[0, \pi]$ each such term satisfies $1 - \cos(hx) \geq 1 - \cos 1$. As $h \leq 2n$, $B(\theta) \geq n \frac{1 - \cos 1}{4n^2} = \frac{1 - \cos 1}{4n}$, and $n \leq 2/x$ gives $B(\theta) \geq \frac{(1 - \cos 1)x}{8}$. Both cases, together with $B(0) = 0$, yield

$$B(\theta) \geq c_b |\theta|. \quad (17)$$

Finite sums and orthogonality of $e^{it\theta}$ express the frequency terms as time differences and give $\frac{1}{2\pi} \int \|\widehat{U}\|^2 = \sum_{t \in \mathbb{Z}} \|u_t\|^2$. For fixed $h \geq 1$ the shifted difference has transform $(e^{-ih\theta} - 1)\widehat{U}(\theta)$, so expanding its squared modulus gives

$$\frac{1}{2\pi} \int (1 - \cos(h\theta)) \|\widehat{U}\|^2 d\theta = \frac{1}{2} \sum_{t \in \mathbb{Z}} \|u_{t+h} - u_t\|^2. \quad (18)$$

For each lag, the zero-extended total variation is $\sum_{t \in \mathbb{Z}} \|u_{t+1} - u_t\| = \|u_1\| + P_T + \|u_T\| \leq P_T + 2M$; the two endpoint terms must be retained. Telescoping $u_{t+h} - u_t = \sum_{r=0}^{h-1} (u_{t+r+1} - u_{t+r})$,

taking norms and summing over t , each adjacent difference appears exactly h times, so $\sum_t \|u_{t+h} - u_t\| \leq h(P_T + 2M)$. Since every difference has norm at most $2M$,

$$\sum_t \|u_{t+h} - u_t\|^2 \leq 2Mh(P_T + 2M). \quad (19)$$

If $h > T$ the two supports are disjoint, so there are no cross terms and exactly

$$\sum_t \|u_{t+h} - u_t\|^2 = 2 \sum_{t=1}^T \|u_t\|^2 \leq 2TM^2. \quad (20)$$

Combining (17), monotone convergence and (18),

$$\frac{1}{2\pi} \int |\theta| \|\widehat{U}\|^2 \leq \frac{1}{2c_b} \sum_{h \geq 1} \frac{1}{h^2} \sum_t \|u_{t+h} - u_t\|^2.$$

Using (19) for $1 \leq h \leq T$ and (20) for $h > T$, together with $\sum_{h \leq T} h^{-1} \leq 1 + \ln T$ and $\sum_{h > T} h^{-2} \leq 1/T$, gives (16). $\square$

Proposition 2. Let $L = \lceil \log_2 T \rceil$, $D = \max\{4/p_-, 1/\beta\}$ and $c_b = \min\{\frac{1-\cos 1}{8}, \frac{2}{\pi^2}\}$. Define

$$\mathfrak{B}_T(U) = \frac{2D}{a_L} \left[M^2 + \frac{M(P_T + 2M)(1 + \ln T) + M^2}{c_b} \right]. \quad (21)$$

Every compressed kernel of Theorem 1 obeys $\mathcal{E}_{\bar{K}_T^\delta}(U) \leq \mathfrak{B}_T(U)/[2(1 - \delta)]$, and for an absolute constant C_B ,

$$\mathfrak{B}_T(U) \leq C_B(M^2 + MP_T)[\ln(eT)]^2[\ln \ln(e^e T)]^2. \quad (22)$$

Proof of Proposition 2. For a scalar sequence u , the energy is bounded by a frequency integral. For $x \in \mathbb{R}^T$ put $X(\theta) = \sum_t x_t e^{it\theta}$, so $u^\top x = \frac{1}{2\pi} \int \operatorname{Re}(\widehat{u}\bar{X})$. Since $f > 0$ almost everywhere, nonnegativity of $|\widehat{u}/\sqrt{f} - \sqrt{f}X|^2$ gives pointwise $\frac{|\widehat{u}|^2}{f} - 2\operatorname{Re}(\widehat{u}\bar{X}) + f|X|^2 \geq 0$; integrating and using the spectral form of K_T^∞ ,

$$2u^\top x - x^\top K_T^\infty x \leq \frac{1}{2\pi} \int \frac{|\widehat{u}|^2}{f} d\theta.$$

Taking the supremum over x and invoking (6), then summing over the d coordinates,

$$\mathcal{E}_{K_T^\infty}(U) \leq \frac{1}{2\pi} \int \frac{\|\widehat{U}(\theta)\|^2}{f(\theta)} d\theta. \quad (23)$$

To bound the reciprocal spectrum, let $L = \lceil \log_2 T \rceil$. Then $\frac{1}{2T} < \lambda_L \leq \frac{1}{T}$. If $|\theta| < 1/T$, Lemma 3 gives $f \geq a_L p - \frac{\lambda_L}{\lambda_L^2 + \theta^2} \geq \frac{p - a_L T}{4}$. If $1/T \leq |\theta| \leq 1$, let $J = J(\theta) \leq L$, so $a_J \geq a_L$ and, by the single-component bound of Lemma 4 together with $2^J > \frac{1}{2|\theta|}$, $f \geq \frac{p - a_L}{4|\theta|}$. If $1 < |\theta| \leq \pi$ then $f \geq \beta$ and $a_L \leq 1$. Combining the three bands,

$$\frac{1}{f(\theta)} \leq \frac{D}{a_L} \left(\frac{1}{T} + |\theta| \right) \quad \text{a.e.} \quad (24)$$

Substituting (24) into (23) and using $\frac{1}{T} \cdot \frac{1}{2\pi} \int \|\widehat{U}\|^2 = \frac{1}{T} \sum_t \|u_t\|^2 \leq M^2$ together with Lemma 7 gives $\mathcal{E}_{K_T^\infty}(U) \leq \mathfrak{B}_T(U)/2$. By (8) and Lemma 1, $\mathcal{E}_{\bar{K}_T^\delta}(U) \leq \mathcal{E}_{K_T^\infty}(U)/(1 - \delta)$, which is the first claim.

For the asymptotic form, Lemma 2 gives $a_L^{-1} \leq (L+e+1)[\ln(L+e+1)]^2$. Put $x = \ln(eT) \geq 1$ and $C_L = e + 2 + 1/\ln 2$; since $L \leq \frac{\ln T}{\ln 2} + 1$ we get $L + e + 1 \leq C_L x$, hence $\ln(L + e + 1) \leq \ln C_L + \ln x$. Writing $y = \ln(e + \ln T) \geq 1$ and noting $\ln x \leq y$ gives $\ln(L + e + 1) \leq (1 + \ln C_L)y$. Finally $M(P_T + 2M) \leq 2(M^2 + MP_T)$ and $x \geq 1$ bound the bracket of (21) by $(1 + 3/c_b)(M^2 + MP_T)x$, so (22) holds with $C_B = 2D C_L(1 + \ln C_L)^2(1 + 3/c_b)$. $\square$

Lemma 8. *Let $G_1, \dots, G_T$ lie in a real Hilbert space, $S_{t-1} = \sum_{s < t} G_s$ and $W_{t-1} = \sum_{s < t} \|G_s\|^2$. The choice $v_t = -S_{t-1}/\max\{\|S_{t-1}\|, \sqrt{W_{t-1}}\}$, with $v_t = 0$ when $S_{t-1} = 0$, satisfies $\|v_t\| \leq 1$ and $\sum_t \langle G_t, v_t - u \rangle \leq 6\sqrt{W_T}$ for every $\|u\| \leq 1$. No bound on $\|G_t\|$ is required.*

Proof of Lemma 8. Write $\gamma_t = \sqrt{W_{t-1}}$ and $\zeta_t = \|G_t\|$.

Fix $S \neq 0$ and $\gamma \geq 0$. At fixed radius $r \in [0, 1]$, Cauchy Schwarz gives $\langle S, v \rangle \geq -\|S\|r$ with equality at $v = -rS/\|S\|$, so the problem reduces to minimizing $-r\|S\| + \frac{\gamma}{2}r^2$ over $r \in [0, 1]$. For $\gamma > 0$ the derivative $-\|S\| + \gamma r$ gives the minimizer $r = \min\{1, \|S\|/\gamma\}$; for $\gamma = 0$ the objective is strictly decreasing, so $r = 1$. This is the stated form of v_t . If $W_{t-1} = 0$ then every past G_s vanishes, so $S_{t-1} = 0$ and no division by zero occurs.

For stability at a fixed regularizer, let $\gamma > 0$, and let v and v^+ minimize $\langle S, z \rangle + \frac{\gamma}{2}\|z\|^2$ and $\langle S + G, z \rangle + \frac{\gamma}{2}\|z\|^2$ on the unit ball. Optimality along the segment from v to any feasible z gives $\langle S + \gamma v, z - v \rangle \geq 0$; taking $z = v^+$, and symmetrically $\langle S + G + \gamma v^+, v - v^+ \rangle \geq 0$, and adding,

$$\gamma\|v - v^+\|^2 \leq \langle G, v - v^+ \rangle \leq \|G\|\|v - v^+\|,$$

so $\|v - v^+\| \leq \|G\|/\gamma$. Both points lie in the unit ball, so also $\|v - v^+\| \leq 2$.

Now define the telescoping potential

$$F_t(S) = \min_{\|v\| \leq 1} \left\{ \langle S, v \rangle + \frac{\gamma_t}{2}\|v\|^2 \right\}.$$

Since $\gamma_{t+1} \geq \gamma_t$, $F_{t+1} \geq F_t$ pointwise. Let v_t^+ attain $F_t(S_t)$ and set $e_t = \langle G_t, v_t \rangle - (F_t(S_t) - F_t(S_{t-1}))$. Using v_t as a candidate for $F_t(S_t)$ gives $e_t \geq 0$; using v_t^+ as a candidate for $F_t(S_{t-1})$ gives $F_t(S_t) - F_t(S_{t-1}) \geq \langle G_t, v_t^+ \rangle$, so

$$e_t \leq \langle G_t, v_t - v_t^+ \rangle \leq \begin{cases} \min\{2\zeta_t, \zeta_t^2/\gamma_t\}, & \gamma_t > 0, \\ 2\zeta_t, & \gamma_t = 0. \end{cases} \quad (25)$$

Summing and using monotonicity of F_t ,

$$\sum_t \langle G_t, v_t \rangle \leq F_{T+1}(S_T) - F_1(S_0) + \sum_t e_t,$$

and $S_0 = 0$, $\gamma_1 = 0$ give $F_1(S_0) = 0$. Any unit-ball u is a candidate in F_{T+1} , so

$$\sum_t \langle G_t, v_t - u \rangle \leq \frac{\sqrt{W_T}}{2}\|u\|^2 + \sum_t e_t. \quad (26)$$

To bound the stability sum, put $\Delta_t = \sqrt{W_{t-1} + \zeta_t^2} - \sqrt{W_{t-1}}$. If $\zeta_t = 0$ then $e_t = \Delta_t = 0$. If $\zeta_t \geq \gamma_t > 0$ or $\gamma_t = 0 < \zeta_t$, then $\Delta_t = \frac{\zeta_t^2}{\sqrt{\gamma_t^2 + \zeta_t^2} + \gamma_t} \geq \frac{\zeta_t}{\sqrt{2} + 1}$, and the bound $e_t \leq 2\zeta_t$ from (25) gives $e_t \leq 2(\sqrt{2} + 1)\Delta_t$. If $0 < \zeta_t < \gamma_t$ then $\frac{\zeta_t^2}{\gamma_t} = \frac{\sqrt{\gamma_t^2 + \zeta_t^2} + \gamma_t}{\gamma_t} \Delta_t \leq (\sqrt{2} + 1)\Delta_t$, and the second bound in (25) gives the same conclusion. Hence $\sum_t e_t \leq 2(\sqrt{2} + 1)\sum_t \Delta_t = 2(\sqrt{2} + 1)\sqrt{W_T}$. Substituting into (26) with $\|u\| \leq 1$,

$$\sum_t \langle G_t, v_t - u \rangle \leq \left(\frac{1}{2} + 2 + 2\sqrt{2}\right)\sqrt{W_T} \leq 6\sqrt{W_T}. \quad \square$$

Lemma 9. Let $|b_t| \leq G$ and $\sum_{t \leq T} b_t^2 \leq V \leq G^2 T$. The magnitude (4) satisfies, for every $r \geq 0$,

$$\sum_{t=1}^T b_t(x_t - r) \leq \varepsilon + 3r\sqrt{V\mathcal{L}(r, T)} + 4rG\mathcal{L}(r, T), \quad (27)$$

where $\mathcal{L}$ is (5), using at most $1 + \lceil 2\log_2(t+1) \rceil$ exponentials and two persistent scalars per round.

Proof of Lemma 9. The analysis uses an infinite potential. Partial fractions give $\pi_i = \frac{1}{i+1} - \frac{1}{i+2}$ and $\sum_{i=0}^N \pi_i = 1 - \frac{1}{N+2}$, so $\sum_{i \geq 0} \pi_i = 1$. Define

$$\Phi_t^\infty = \varepsilon_0 \sum_{i \geq 0} \pi_i e^{\eta_i s_t - \eta_i^2 q_t}, \quad x_t^\infty = \varepsilon_0 \sum_{i \geq 0} \pi_i \eta_i e^{\eta_i s_t - \eta_i^2 q_t}. \quad (28)$$

Because $q_t \geq 0$ and $0 < \eta_i \leq \frac{1}{2G}$, the i -th terms are at most $\pi_i e^{|s_t|/(2G)}$ and $\pi_i e^{|s_t|/(2G)}/(2G)$ respectively, so both series converge absolutely for finite t , with $\Phi_1^\infty = \varepsilon_0$.

For the wealth inequality, let $|u| \leq \frac{1}{2}$ and put $F(u) = \ln(1-u) + u + u^2$, whose derivative is $\frac{u(1-2u)}{1-u}$. This is nonnegative on $[0, \frac{1}{2}]$ and nonpositive on $[-\frac{1}{2}, 0]$, and $F(0) = 0$, so $F \geq 0$ on the whole interval, i.e.

$$e^{-u-u^2} \leq 1-u, \quad |u| \leq \frac{1}{2}. \quad (29)$$

Since $|\eta_i b_t| \leq \frac{1}{2}$, applying (29) termwise,

$$e^{\eta_i(s_t - b_t) - \eta_i^2(q_t + b_t^2)} \leq e^{\eta_i s_t - \eta_i^2 q_t} (1 - \eta_i b_t),$$

and absolute convergence permits summing, giving $\Phi_{t+1}^\infty \leq \Phi_t^\infty - b_t x_t^\infty$ and hence $\sum_{t \leq T} b_t x_t^\infty \leq \varepsilon_0 - \Phi_{T+1}^\infty$.

All terms are nonnegative, so the truncation can be charged directly. Thus $\Delta_t = x_t^\infty - x_t \geq 0$. From $|s_t| \leq G(t-1)$ and $i > I_t$ we get $\eta_i \leq \eta_{I_t+1} \leq \frac{1}{4G(t+1)^2}$, hence $\eta_i s_t - \eta_i^2 q_t \leq \eta_i |s_t| \leq \frac{t-1}{4(t+1)^2} \leq \frac{1}{4}$; and $e^{1/4} = \sum_{k \geq 0} \frac{4^{-k}}{k!} \leq \sum_{k \geq 0} 4^{-k} = \frac{4}{3}$. Therefore

$$\begin{aligned} \Delta_t &= \varepsilon_0 \sum_{i > I_t} \pi_i \eta_i e^{\eta_i s_t - \eta_i^2 q_t} \leq \frac{\varepsilon_0}{3G(t+1)^2} \sum_{i > I_t} \pi_i \\ &\leq \frac{\varepsilon_0}{3G(t+1)^2}. \end{aligned} \quad (30)$$

Since x^{-2} decreases, $\sum_{t \geq 1} (t+1)^{-2} \leq 1$, so $|b_t| \leq G$ gives

$$\begin{aligned} \sum_{t \leq T} b_t x_t &\leq \varepsilon_0 - \Phi_{T+1}^\infty + G \sum_{t \leq T} \Delta_t \\ &\leq \frac{4}{3} \varepsilon_0 - \Phi_{T+1}^\infty = \varepsilon - \Phi_{T+1}^\infty. \end{aligned} \quad (31)$$

Note that (28) is defined along the realized feedback sequence b_t , so (31) holds path by path against adaptive environments. For $r = 0$ the potential is nonnegative and (31) already gives (27); assume $r > 0$ below.

To retain one prior component, put $s = -\sum_t b_t$ and $q = \sum_t b_t^2$. For any $i \geq 0$, (31) gives $\sum_t b_t(x_t - r) \leq \varepsilon + rs - \varepsilon_0 \pi_i e^{\eta_i s - \eta_i^2 q}$. The function $F(z) = rz - \varepsilon_0 \pi_i e^{\eta_i z - \eta_i^2 q}$ has strictly negative second derivative and derivative tending to $r > 0$ as $z \rightarrow -\infty$ and to $-\infty$ as $z \rightarrow +\infty$, so its unique stationary point is a global maximum; solving $\eta_i z = \ln \frac{r}{\varepsilon_0 \pi_i \eta_i} + \eta_i^2 q$ and substituting back,

$$\sum_t b_t(x_t - r) \leq \varepsilon + r\eta_i V + \frac{r}{\eta_i} \left(\ln \frac{r}{\varepsilon_0 \pi_i \eta_i} - 1 \right). \quad (32)$$

To select a rate from the prior, write $I = \lceil \frac{1}{2} \log_2 T \rceil$ and $L = \mathcal{L}(r, T)$, as in (5); note I appears only here and never in the algorithm. For $0 \leq i \leq I$ we have $\pi_i \geq \pi_I$ and $\eta_i \geq \eta_I$, so $\frac{r}{\varepsilon_0 \pi_i \eta_i} \leq \frac{2rG}{\varepsilon_0} 2^I (I+1)(I+2)$, and since $\ln z - 1 \leq 1 + \ln(1+z)$ for $z > 0$, the bracket in (32) is at most L . If $V = 0$ all b_t vanish and the claim is trivial, so let $V > 0$. If $\sqrt{L/V} \geq \frac{1}{2G}$, take $i = 0$; then $V \leq 4G^2L$ and $\eta_0 V + L/\eta_0 = \frac{V}{2G} + 2GL \leq 4GL$. Otherwise put $\eta_\star = \sqrt{L/V}$; from $L \geq 1$ and $V \leq G^2T$ we get $\eta_\star \geq \frac{1}{G\sqrt{T}}$ while $\eta_I \leq \frac{1}{2G\sqrt{T}}$, and since consecutive rates differ by a factor 2 there is $0 \leq i \leq I$ with $\eta_\star/2 < \eta_i \leq \eta_\star$, whence $\eta_i V + L/\eta_i \leq 3\sqrt{VL}$. Substituting either branch into (32) and adding nonnegative terms to unify gives (27). The analysis selects a prior component, but the bounds in (30) and (31) have already charged the cumulative effect of every component the algorithm did not evaluate, which is what makes the argument valid. The running algorithm stores only s_t and q_t and regenerates η_i, π_i on the fly. $\square$

Proof of Theorem 2. Fix any evaluation horizon T and interpret the algorithm in $\mathcal{H}_{\bar{K}_T^\delta}^d$ with $G_t = \bar{k}^\delta(t, \cdot) g_t$ and $S_{t-1} = \sum_{s < t} G_s$. The reproducing property and $\bar{k}^\delta(t, t) = 1$ give $\|G_t\|^2 = \|g_t\|^2$, $S_{t-1}(t) = h_t$ and $\|S_{t-1}\|^2 = N_t$, so the implemented direction update is exactly the rule of Lemma 8 with $W_{t-1} = V_{t-1}$. Since $\bar{k}^\delta$ does not depend on T and every maintained quantity uses only the current round and past feedback, this retrospective choice of T changes no prediction the algorithm ever made. Moreover $|b_t| = |\langle G_t, v_t \rangle| \leq \|G_t\| \|v_t\| \leq \|g_t\| \leq G$ and $\sum_t b_t^2 \leq V_T \leq G^2T$.

Convexity gives

$$R_T(U) \leq \sum_t \langle g_t, w_t - u_t \rangle = \sum_t \langle G_t, x_t v_t - U \rangle.$$

If $r > 0$ then $\|U/r\|_{\mathcal{H}^d} = 1$ and expanding termwise,

$$\sum_t \langle G_t, x_t v_t - U \rangle = \sum_t b_t(x_t - r) + r \sum_t \langle G_t, v_t - U/r \rangle.$$

Lemma 9 bounds the first sum by $\varepsilon + 3r\sqrt{V_T \mathcal{L}(r, T)} + 4rG\mathcal{L}(r, T)$ and Lemma 8 bounds the second by $6r\sqrt{V_T}$; since $\mathcal{L} \geq 1$, this gives the stated RKHS-rate bound. If $r = 0$ then strict positive definiteness forces $U = 0$ and (31) gives $R_T(U) \leq \varepsilon$.

For the substitution, Theorem 1 and the frequency bound in the proof of Proposition 2 give respectively

$$r^2 \leq \frac{\mathcal{E}_{K_T^Q}(U)}{(1-\delta)c_\star}, \quad r^2 \leq \frac{\mathfrak{B}_T(U)}{2(1-\delta)},$$

hence the displayed RKHS rate follows after replacing r by the smaller kernel-energy/path-length upper bound. Combining with (22) yields (12). $\square$

Lemma 10. For all $n \geq 1$ and $0 < \rho < 1$, $c_H \psi(n\mu) \leq n^{-1} (h^{(n)})^\top R_{\rho, 2n} h^{(n)} \leq C_H \psi(n\mu)$.

Proof of Lemma 10. On an orthonormal basis $(e_\ell)_{\ell \in \mathbb{Z}}$ of $\ell^2(\mathbb{Z})$ set

$$\phi_t = \sqrt{1 - \rho^2} \sum_{\ell \leq t} \rho^{t-\ell} e_\ell.$$

Since $(1 - \rho^2) \sum_{k \geq 0} \rho^{2k} = 1$ the series converges and $\|\phi_t\| = 1$, and for $s \leq t$,

$$\langle \phi_s, \phi_t \rangle = \rho^{t-s} (1 - \rho^2) \sum_{k \geq 0} \rho^{2k} = \rho^{t-s},$$

so the Gram matrix of $\phi_1, \dots, \phi_{2n}$ is $R_{\rho, 2n}$. Hence $v_n := (h^{(n)})^\top R_{\rho, 2n} h^{(n)} = \|\sum_t h_t^{(n)} \phi_t\|^2$.

Keeping only the coordinates $\ell = n + 1, \dots, 2n$, which receive contributions solely from the negative half of the probe, the absolute coefficient at coordinate ℓ is $\sqrt{\frac{1-\rho^2}{2n} \frac{1-\rho^{2n-\ell+1}}{1-\rho}}$, so with $k = 2n - \ell + 1$,

$$v_n \geq \frac{1}{2n} \frac{1+\rho}{1-\rho} \sum_{k=1}^n (1-\rho^k)^2. \quad (33)$$

Suppose $\lambda \leq 1$ and $n\lambda \leq 1$. For $1 \leq k \leq n$, $1 - e^{-\lambda k} = \int_0^{\lambda k} e^{-x} dx \geq e^{-1} \lambda k$, while $1 - e^{-\lambda} \leq \lambda$ gives $\frac{1+\rho}{1-\rho} \geq \frac{1}{\lambda}$. Since $\sum_{k \leq n} k^2 \geq n^3/3$, (33) yields $v_n \geq \frac{e^{-2}}{6} \lambda n^2$. For the upper bound, $\sum_s h_s^{(n)} = 0$ lets us write $v_n = (h^{(n)})^\top (R_{\rho, 2n} - \mathbf{1}\mathbf{1}^\top) h^{(n)}$; each entry of that matrix has modulus at most $1 - e^{-\lambda|s-t|} \leq \lambda|s-t| \leq 2\lambda n$, and $(\sum_s |h_s^{(n)}|)^2 = 2n$, so $v_n \leq 4\lambda n^2$. Dividing by n and using $\mu = \lambda$, $\psi(n\lambda) = n\lambda$ and $4 \leq 2e$ gives the claim.

Now suppose $n\mu \geq 1$. (If $\lambda \leq 1$ the previous case failing means $n\lambda > 1$; if $\lambda > 1$ then $\mu = 1$ and $n \geq 1$.) For any $x \in \mathbb{R}^{2n}$, using $2|ab| \leq a^2 + b^2$,

$$\begin{aligned} x^\top R_{\rho, 2n} x &\leq \sum_s x_s^2 \sum_t \rho^{|s-t|} \\ &\leq \left(1 + 2 \sum_{k \geq 1} \rho^k\right) \|x\|^2 = \frac{1+\rho}{1-\rho} \|x\|^2. \end{aligned} \quad (34)$$

If $\lambda \leq 1$ then $1 - e^{-\lambda} \geq e^{-1} \lambda$, so the ratio is at most $2e/\lambda$; if $\lambda > 1$ then $1 - e^{-\lambda} \geq 1 - e^{-1} \geq e^{-1}$, so it is at most $2e$. Either way $v_n \leq 2e/\mu$. For the lower bound, the set $\{\lceil n/2 \rceil, \dots, n\}$ contains at least $n/2$ integers, each with $\lambda k \geq \frac{1}{2}$, and $\frac{1+\rho}{1-\rho} \geq \frac{1}{\mu}$ in both regimes, so (33) gives $v_n \geq \frac{1}{2n\mu} \cdot \frac{n}{2} (1 - e^{-1/2})^2 = \frac{(1-e^{-1/2})^2}{4\mu}$. Dividing by n and using $\psi(n\mu) = 1/(n\mu)$ completes the case, and the two cases exhaust all $n \geq 1$, $\lambda > 0$. $\square$

Lemma 11. For $Y_j^\infty = 2^{-j} (h^{(2^j)})^\top K_{2^j+1}^\infty h^{(2^j)}$ one has $c_H a_j \leq Y_j^\infty \leq 103 C_H a_j$; and if $L \geq 1$ and $p = \lceil L/2 \rceil$, then $a_L \leq a_j \leq 8a_L$ for $p \leq j \leq L$.

Proof of Lemma 11. Every entry of the infinite mixture converges absolutely, so the finite quadratic form may be interchanged with the series, and Lemma 10 applied with $\mu_k = \lambda_k$ and $\psi(2^{j-k}) = 2^{-|j-k|}$ gives

$$c_H \sum_{k \geq 0} a_k 2^{-|j-k|} \leq Y_j^\infty \leq C_H \sum_{k \geq 0} a_k 2^{-|j-k|}. \quad (35)$$

Retaining the term $k = j$ shows the middle sum is at least a_j . For the upper bound, split at $k = j$: Lemma 2 gives $\sum_{k \leq j} a_k 2^{-(j-k)} = 2^{-j} \sum_{k \leq j} 2^k a_k \leq 102 a_j$, and monotonicity of the weights gives $\sum_{k > j} a_k 2^{-(k-j)} \leq a_j \sum_{r \geq 1} 2^{-r} = a_j$; adding yields the first claim.

For the second, $a_j \geq a_L$ by monotonicity, and

$$\begin{aligned} \frac{a_j}{a_L} &\leq \frac{a_p}{a_L} \leq \frac{w(p+e)}{w(L+e+1)} \\ &= \frac{L+e+1}{p+e} \left(\frac{\ln(L+e+1)}{\ln(p+e)} \right)^2. \end{aligned}$$

Put $\varrho = (L+e+1)/(p+e)$; then $\varrho \geq 1$ since $p \leq L$, and $\varrho \leq 2$ since $2p \geq L$ and $e \geq 1$. For $x \geq e$ and $\varrho \geq 1$ the function $\varrho \ln x - \ln(\varrho x)$ has derivative $\ln x - 1/\varrho \geq 0$ in ϱ and vanishes at $\varrho = 1$, so $\ln(\varrho x)/\ln x \leq \varrho$; taking $x = p+e$ bounds the display by $\varrho^3 \leq 8$. $\square$

Lemma 12. For real r and integers $p \leq L$, $\sum_{j=p}^L 2^{-|j-r|} \leq 5 \max_{p \leq j \leq L} 2^{-|j-r|}$.

Proof of Lemma 12. If $r < p$ the maximum is at $j = p$ and the $(p + k)$ -th term is 2^{-k} times the maximum, so the sum is at most 2 times the maximum; the case $r > L$ is symmetric. Otherwise $p \leq r \leq L$; choose the integer j_0 in $[p, L]$ nearest r , so $|j_0 - r| \leq \frac{1}{2}$ and its term is the maximum. For $k \geq 1$, the triangle inequality gives $|j_0 \pm k - r| - |j_0 - r| \geq k - 2|j_0 - r| \geq k - 1$, so the terms on each side of the peak sum to at most $\sum_{k \geq 1} 2^{1-k} = 2$ times the peak. Adding the peak itself gives the factor $1 + 2 + 2 = 5$; discarding terms outside $[p, L]$ only decreases the sum, which also covers a peak at an endpoint. $\square$

C State lower-bound statements

The lower bounds use two architectural assumptions.

Assumption 1 (Positive stationary exponential approximator). The surrogate has the form

$$\tilde{k}_T(s, t) = \sum_{i=1}^m b_i \rho_i^{|s-t|},$$

with $b_i > 0$ and $\rho_i \in (0, 1)$; the parameters may depend on T .

Assumption 2 (Linear causal realization). The convolution is generated by $q_1 = 0$, $q_{t+1} = A_t q_t + B_t g_t$, $h_t = C_t q_t$, with $q_t \in \mathbb{R}^r$. The matrices may depend on t , T and the kernel, but not on the input.

For the scale-width bound, let $h^{(n)} \in \mathbb{R}^{2n}$ be the unit Haar vector, let $\lambda = -\ln \rho$, $\mu = \min\{\lambda, 1\}$ and $\psi(x) = \min\{x, x^{-1}\}$. Set $c_H = \min\{e^{-2}/6, (1 - e^{-1/2})^2/4\}$, $C_H = 2e$ and $\kappa_H = c_H^2/(4120C_H^2)$. For $2^{L+1} \leq T$, define

$$\mathcal{W}_H(K_T; p, L) = \frac{\sum_{j=p}^L Y_j(K_T)}{\max_{p \leq j \leq L} Y_j(K_T)}, \quad Y_j(K_T) = 2^{-j} (h^{(2^j)})^\top K_T h^{(2^j)}, \quad (36)$$

where the Haar probe is padded to length T .

Theorem 2. Let Assumption 1 hold and suppose $\gamma_- K_T \preceq \tilde{K}_T \preceq \gamma_+ K_T$ for a strictly positive definite K_T . Then

$$m \geq \frac{c_H}{5C_H} \frac{\gamma_-}{\gamma_+} \mathcal{W}_H(K_T; p, L). \quad (37)$$

For $K_T = K_T^\infty$, $T \geq 4$, $L = \lfloor \log_2 T \rfloor - 1$ and $p = \lceil L/2 \rceil$, this is $m \geq \kappa_H (\gamma_-/\gamma_+) N_L = \Omega(\log T)$ at fixed distortion. Against K_T^Q , the ratio is replaced by $ac_*/(bC_*)$ when $aK_T^Q \preceq \tilde{K}_T \preceq bK_T^Q$.

Theorem 3. Let $n = \lfloor T/2 \rfloor$ and let Assumption 2 hold. If $k(s, t) = \sum_{i=1}^m b_i \rho_i^{|s-t|}$ with positive weights and distinct rates, then $r \geq \min\{m, n\}$. For k^∞ , exact realization requires $r \geq n$. With d -dimensional inputs, both bounds are multiplied by d .

Corollary 1. Within Assumptions 1–2, fixed-distortion preservation of the Loewner geometry of K_T^∞ requires $\Omega(d \log T)$ state coordinates, matched by Theorem 1 up to constants; exact realization requires $\Omega(dT)$.

Proof of Theorem 2. Set, for each component of the approximator,

$$\mu_i = \min\{-\ln \rho_i, 1\} \in (0, 1], \quad r_i = -\log_2 \mu_i \geq 0,$$

and for $p \leq j \leq L$ define

$$B_j = \sum_{i=1}^m b_i 2^{-|j-r_i|},$$

$$\tilde{Y}_j = 2^{-j} (h^{(2^j)})^\top \tilde{K}_T h^{(2^j)},$$

each probe being padded with zeros into $\mathbb{R}^T$. Lemma 10 with $\psi(2^j \mu_i) = 2^{-|j-r_i|}$ gives $c_H B_j \leq \tilde{Y}_j \leq C_H B_j$, while applying the Loewner hypothesis to the same probe gives $\gamma_- Y_j(K_T) \leq \tilde{Y}_j \leq \gamma_+ Y_j(K_T)$. Combining,

$$\frac{\gamma_-}{C_H} Y_j(K_T) \leq B_j \leq \frac{\gamma_+}{c_H} Y_j(K_T). \quad (38)$$

Let $Y_{\max} = \max_{p \leq j \leq L} Y_j(K_T) > 0$. For each i pick j_i maximizing $2^{-|j-r_i|}$ over the index range; positivity of the weights gives $b_i 2^{-|j_i-r_i|} \leq B_{j_i} \leq \frac{\gamma_+}{c_H} Y_{\max}$, so Lemma 12 gives $\sum_{j=p}^L b_i 2^{-|j-r_i|} \leq \frac{5\gamma_+}{c_H} Y_{\max}$. Summing over i , exchanging the two finite sums and using the left half of (38),

$$\frac{\gamma_-}{C_H} \sum_{j=p}^L Y_j(K_T) \leq \sum_{j=p}^L B_j \leq \frac{5m\gamma_+}{c_H} Y_{\max},$$

and dividing by $Y_{\max} > 0$ gives (37). Note this argument requires nothing of the target beyond strict positive definiteness.

For K_T^∞ , Lemma 11 gives $Y_j(K_T^\infty) \geq c_H a_j \geq c_H a_L$ and $Y_j(K_T^\infty) \leq 103 C_H a_j \leq 824 C_H a_L$ for $p \leq j \leq L$, so $\mathcal{W}_H(K_T^\infty; p, L) \geq \frac{c_H}{824 C_H} N_L$; substituting into (37) and using $\kappa_H = c_H^2 / (4120 C_H^2)$ gives the second claim, and $N_L \geq L/2$ with $L \geq \log_2 T - 2$ makes it $\Omega(\log T)$. Finally Theorem 1 gives $K_T^\infty / C_\star \preceq K_T^Q \preceq K_T^\infty / c_\star$, so a hypothesis stated against K_T^Q implies one against K_T^∞ with $\gamma_- = a/C_\star$ and $\gamma_+ = b/c_\star$. $\square$

Proof of Theorem 3. Set $g_{n+1} = \dots = g_T = 0$ and collect $g_1, \dots, g_n$ into $g^- \in \mathbb{R}^n$. Finitely many substitutions in the recursion of Assumption 2 show $q_{n+1} = M g^-$ for some $M \in \mathbb{R}^{r \times n}$; and since the later inputs vanish, $h_{n+1}, \dots, h_{2n}$ equal $O q_{n+1}$ for some $O \in \mathbb{R}^{n \times r}$ whose $(i+1)$ -st row block is $C_{n+i+1} A_{n+i} \dots A_{n+1}$. The past-to-future map is therefore OM , whose range lies in the range of O , so its rank is at most r . Reversing the order of the past inputs is a right multiplication by a permutation and does not change the rank, so writing $k_\tau = k(t, t - \tau)$ the required map is the Hankel matrix $H_n = (k_{i+j+1})_{i,j=0}^{n-1}$, and every exact realization obeys $r \geq \text{rank } H_n$.

We use the fact that a nonzero real polynomial of degree at most k has at most k distinct real roots; by induction on k , a nonzero constant has no root, and if $p(a) = 0$ then applying $z^j - a^j = (z - a) \sum_{\ell < j} z^{j-1-\ell} a^\ell$ to $p(z) - p(a)$ factors $p(z) = (z - a)q(z)$ with $\deg q \leq k - 1$.

With $v(\rho) = (1, \rho, \dots, \rho^{n-1})^\top$, a direct computation gives $H_n = \sum_{i=1}^m b_i \rho_i v(\rho_i) v(\rho_i)^\top$. If $m \geq n$, then for nonzero x the polynomial $p_x(z) = \sum_{j < n} x_{j+1} z^j$ is not identically zero and has at most $n - 1$ roots, so some ρ_i among the n or more distinct values satisfies $p_x(\rho_i) \neq 0$, whence $x^\top H_n x = \sum_i b_i \rho_i [p_x(\rho_i)]^2 > 0$ and $\text{rank } H_n = n$. If $m < n$, let V have columns $v(\rho_i)$ and let W be its top $m \times m$ block; $W^\top x = 0$ would make a polynomial of degree at most $m - 1$ vanish at m distinct points, forcing $x = 0$, so W is invertible and V has column rank m . With $D = \text{diag}(b_i \rho_i) \succ 0$ we have $H_n = V D V^\top$, whose kernel equals that of $V^\top$, so $\text{rank } H_n = m$. Together these give $r \geq \min\{m, n\}$.

For the infinite mixture, $H_n^\infty = \sum_{j \geq 0} a_j \rho_j v(\rho_j) v(\rho_j)^\top$ converges entrywise because $\sum_j a_j = 1$ and $\rho_j \in (0, 1)$. The $\rho_j = e^{-2^{-j}}$ are pairwise distinct, and for nonzero x the polynomial p_x has finitely many roots, so at least one term is strictly positive and the rest nonnegative; hence $x^\top H_n^\infty x > 0$ and the rank is n .

For d -dimensional inputs both the past input space and the future output space have dimension nd , and the map still factors through $\mathbb{R}^r$. Ordering coordinates by time makes the required matrix $H_n \otimes I_d$; ordering by coordinate makes it block diagonal with d copies of H_n acting on disjoint subspaces, so the ranks add and the total is $d \operatorname{rank} H_n$. $\square$

Proof of Corollary 1. Merging equal decay rates leaves the mixture unchanged, so m counts distinct rates. Set $c = \kappa_H \gamma_- / \gamma_+$. The two lower-bound theorems give $r \geq d \min\{m, n\} \geq dcN_L = \Omega(d \log T)$, since $N_L \leq n$. The matching upper bound is Theorem 1; exact realization is the second part of Theorem 3. $\square$

Theorem 4. *Let $u_t = v$ for all t , $M = \|v\|$, $L_T = \lceil \log_2 T \rceil$, $\Lambda_T = \ln(L_T + e)$ and $C_S = 408e + 1$. Then $C_S^{-1} M^2 \Lambda_T \leq \mathcal{E}_{K_T^\infty}(U) \leq 2M^2 \Lambda_T$, and consequently*

$$\frac{M^2 \Lambda_T}{(1 + \delta) C_S} \leq \mathcal{E}_{\bar{K}_T^\delta}(U) \leq \frac{2M^2 \Lambda_T}{1 - \delta}. \quad (39)$$

Proof of Theorem 4. If $M = 0$ everything vanishes, so let $M > 0$; expanding the trace coordinate-wise gives $\mathcal{E}_K(U) = M^2 \mathbf{1}^\top K^{-1} \mathbf{1}$, and it suffices to treat the scalar constant sequence $\mathbf{1}$.

For the upper bound, take $j \geq L_T$, so $T\lambda_j \leq 1$. The explicit inverse of an AR kernel gives $\mathbf{1}^\top R_{\rho_j, T}^{-1} \mathbf{1} = 1 + (T - 1) \frac{1 - \rho_j}{1 + \rho_j} \leq 1 + T\lambda_j \leq 2$. Applying Cauchy Schwarz in the inner product $\langle a, b \rangle_{R_j} = a^\top R_j b$,

$$(\mathbf{1}^\top x)^2 = \langle R_j^{-1} \mathbf{1}, x \rangle_{R_j}^2 \leq (\mathbf{1}^\top R_j^{-1} \mathbf{1})(x^\top R_j x) \leq 2x^\top R_j x,$$

so $R_j \succeq \frac{1}{2} \mathbf{1} \mathbf{1}^\top$. Discarding the other nonnegative terms and using the tail identity, $K_T^\infty \succeq \frac{1}{2\Lambda_T} \mathbf{1} \mathbf{1}^\top$. Putting $q = \mathbf{1}^\top (K_T^\infty)^{-1} \mathbf{1} > 0$ and testing this comparison at $x = (K_T^\infty)^{-1} \mathbf{1}$ gives $q \geq q^2 / (2\Lambda_T)$, i.e. $q \leq 2\Lambda_T$.

For the lower bound, each entry of R_j is at most 1, so $\mathbf{1}^\top R_j \mathbf{1} \leq T^2$; and (34) with $\lambda_j \leq 1$ gives $\mathbf{1}^\top R_j \mathbf{1} \leq T \frac{1 + \rho_j}{1 - \rho_j} \leq 2eT2^j$. Splitting the mixture at $j = L_T$,

$$\mathbf{1}^\top K_T^\infty \mathbf{1} \leq 2eT \sum_{j < L_T} 2^j a_j + T^2 \sum_{j \geq L_T} a_j.$$

If $L_T \geq 1$, adding a nonnegative term and applying Lemma 2 gives $\sum_{j < L_T} 2^j a_j \leq 102 \cdot 2^{L_T} a_{L_T}$, which also holds for the empty sum. Since $2^{L_T} \leq 2T$ and $a_{L_T} \leq \sum_{j \geq L_T} a_j = 1/\Lambda_T$,

$$\mathbf{1}^\top K_T^\infty \mathbf{1} \leq 408eT^2 a_{L_T} + \frac{T^2}{\Lambda_T} \leq \frac{C_S T^2}{\Lambda_T}. \quad (40)$$

Applying Cauchy Schwarz in the K_T^∞ inner product to the pair $(K_T^\infty)^{-1} \mathbf{1}$ and $\mathbf{1}$ gives

$$T^2 \leq [\mathbf{1}^\top (K_T^\infty)^{-1} \mathbf{1}] [\mathbf{1}^\top K_T^\infty \mathbf{1}],$$

so dividing by (40) yields $\mathbf{1}^\top (K_T^\infty)^{-1} \mathbf{1} \geq \Lambda_T / C_S$.

Finally (8) with Lemma 1 gives $\mathcal{E}_{K_T^\infty}(U) / (1 + \delta) \leq \mathcal{E}_{\bar{K}_T^\delta}(U) \leq \mathcal{E}_{K_T^\infty}(U) / (1 - \delta)$, which is (39). $\square$